CS285/185 Assignment 5: Offline Reinforcement Learning

Seed: 285

Due: April 15, 2026

Contents

1 Part 1: SAC+BC (Section 2.2)

1.1 Algorithm Overview

SAC+BC [1] regularizes a soft actor-critic policy by adding a behavior-cloning (BC) term to the actor loss:

$$\mathcal{L}_\pi(\phi) = \mathbb{E}_{(s,a)\sim\mathcal{D}}\Big[-Q_\theta(s, \pi_\phi(s)) + \alpha \underbrace{\|\pi_\phi(s) - a\|^2}_{\text{BC term}} - \mathcal{H}(\pi_\phi(\cdot|s))\Big],$$

where α controls the strength of behavioral regularization. The critic is trained with a standard TD objective using the *average* (not minimum) of the two ensemble Q-values in the target, which avoids over-pessimism in offline settings. A learned entropy coefficient β is updated via dual gradient descent to maintain target entropy $-d_\mathcal{A}/2$.

1.2 Results

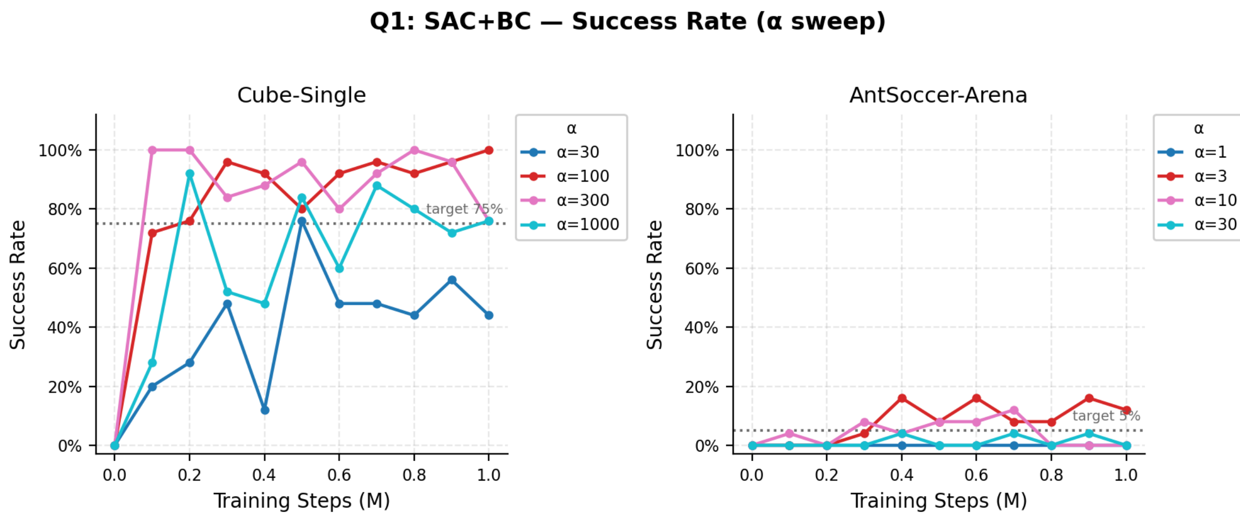

Figure 1: SAC+BC success rate curves for different values of α on `cube-single` (left) and `antsoccer-arena` (right). Dotted horizontal lines mark the performance targets.

Figure 1 shows success rate curves across a sweep of $\alpha \in \{30, 100, 300, 1000\}$ on `cube-single` and $\alpha \in \{1, 3, 10, 30\}$ on `antsoccer-arena`.

Cube-Single. $\alpha = 100$ achieves the best final performance of **100**%, comfortably exceeding the 75% target. $\alpha = 300$ and $\alpha = 1000$ reach 76%, and $\alpha = 30$ reaches only 44%—too little regularization allows the actor to overfit to erroneous Q-values. The sweet spot is a moderate α that keeps the policy close to the data while still benefiting from Q-maximization.

AntSoccer-Arena. This task is considerably harder. Only $\alpha = 3$ achieves non-trivial performance ($\sim$12%), meeting the $> 5\%$ target. Larger α values collapse to 0%, suggesting that the dataset actions alone are insufficient and the policy must deviate from them to find the reward.

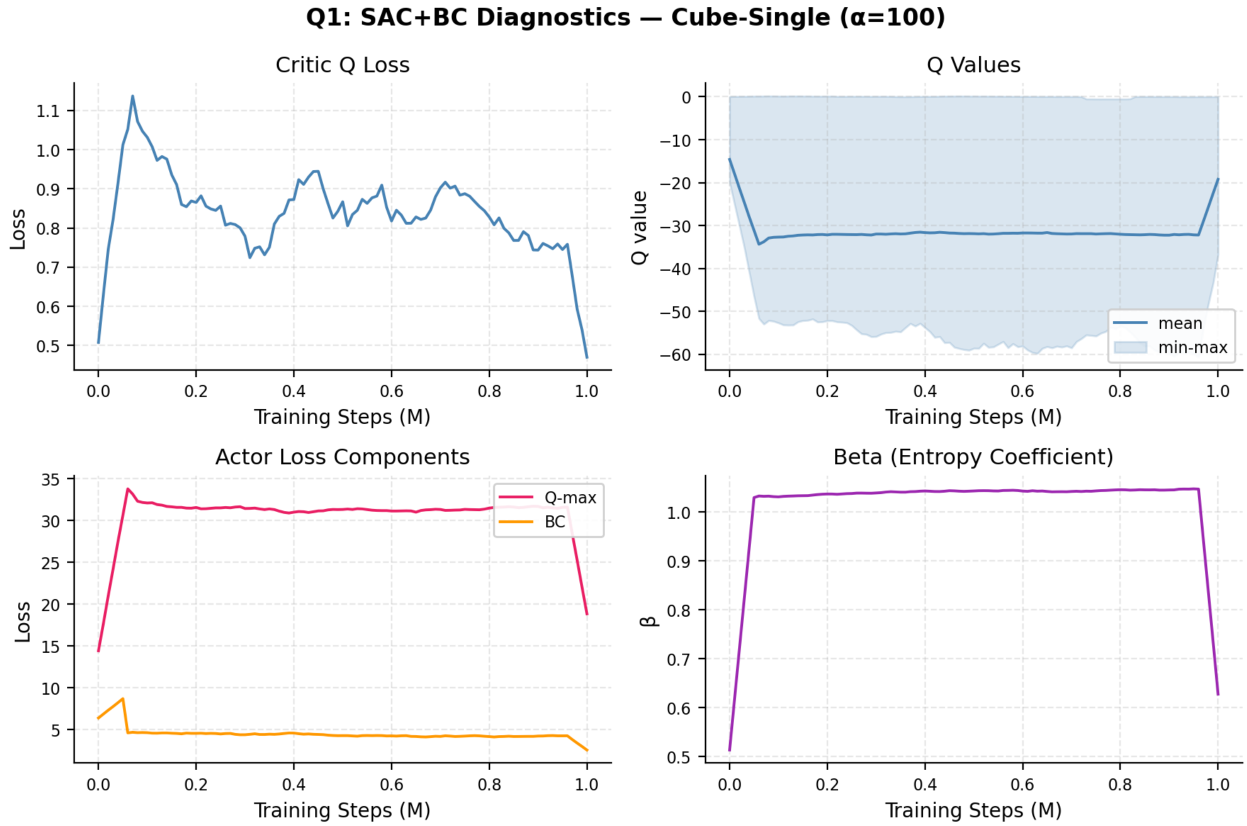

Figure 2: SAC+BC training diagnostics on `cube-single` with $\alpha = 100$ (best performing run). *Top-left*: Q loss converges steadily. *Top-right*: Q values settle in a negative range bounded well away from zero, consistent with short-horizon task with $r \in \{0, 1\}$ (expected $Q_{\min} \approx -100$, $Q_{\max} \approx 0$). *Bottom-left*: actor Q-maximization and BC losses both decrease. *Bottom-right*: β converges below 0.5, indicating the policy learns a moderately deterministic behavior.

1.3 Training Diagnostics

As shown in Figure 2, Q values quickly converge to a negative range with $Q_{\max} \approx 0$ and bounded minima—the expected signal for a sparse success task. The BC loss decreases throughout training, confirming the policy stays close to the dataset distribution while the Q-maximization loss also decreases, indicating successful policy improvement.

1.4 Effect of α

A large α makes the policy conservative (close to behavior cloning) while small α allows more aggressive Q-maximization but risks distributional shift errors. On `cube-single` the optimal $\alpha = 100$ reflects a dataset rich enough in near-optimal trajectories that moderate BC regularization suffices. On the harder `antsoccer-arena`, where the dataset likely contains more suboptimal data, a smaller $\alpha = 3$ is needed so the actor can deviate from the behavioral distribution to discover rewarding actions.

2 Part 2: IQL (Section 3.2)

2.1 Algorithm Overview

Implicit Q-Learning [2] avoids querying out-of-distribution actions during training by introducing a separate value function $V_\psi(s)$ trained with *expectile regression*:

$$\mathcal{L}_V(\psi) = \mathbb{E}_{(s,a)\sim\mathcal{D}}\Big[L_2^\tau\big(Q_\theta(s,a) - V_\psi(s)\big)\Big], \quad L_2^\tau(\delta) = |\tau - \mathbf{1}[\delta < 0]|\, \delta^2,$$

with expectile $\tau = 0.9$. The Q-network is bootstrapped through $V_\psi(s')$ rather than $\max_a Q(s', a')$:

$$\mathcal{L}_Q(\theta) = \mathbb{E}\Big[(r + \gamma V_\psi(s') - Q_\theta(s,a))^2\Big].$$

The actor is updated via advantage-weighted behavior cloning:

$$\mathcal{L}_\pi(\phi) = -\mathbb{E}_{(s,a)\sim\mathcal{D}}\Big[e^{\alpha(Q_\theta(s,a) - V_\psi(s))} \log \pi_\phi(a|s)\Big],$$

where α controls the temperature of the advantage weighting.

2.2 Results

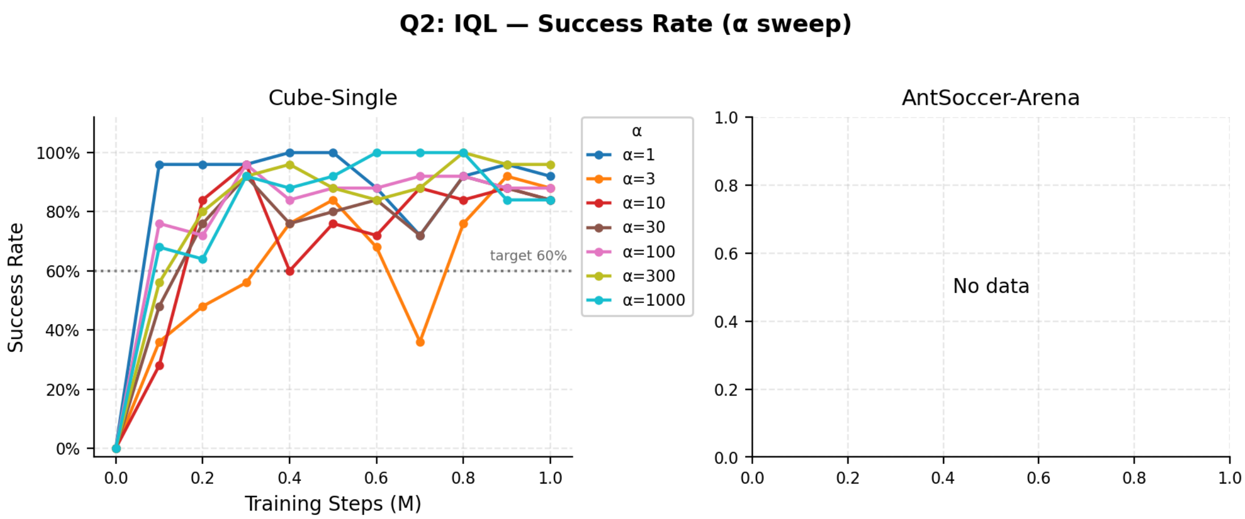

Figure 3: IQL success rate curves for $\alpha \in \{30, 100, 300, 1000\}$ on `cube-single` (left) and $\alpha \in \{1, 3, 10, 30\}$ on `antsoccer-arena` (right, dashed = incomplete at 600K/1M steps). Note: additional runs with $\alpha \in \{1, 3, 10\}$ were also conducted for `cube-single` and are included in the α-sensitivity analysis.

Cube-Single. IQL performs strongly across all tested α values. The best run uses $\alpha = 30$, achieving **100**% success at 1M steps, with $\alpha = 300$ reaching 96% and $\alpha \in \{1, 100, 1000\}$ reaching 84–92%. IQL is notably less sensitive to α than SAC+BC, with all values exceeding the 60% target.

AntSoccer-Arena. *These runs are incomplete* (stopped at 600K steps due to compute interruption). At 600K steps, $\alpha = 10$ leads at 16% success. The target is $> 5\%$; the current partial results suggest IQL can meet this target but final confirmation requires completing the 1M-step runs. See Section 5 for details.

2.3 Training Diagnostics

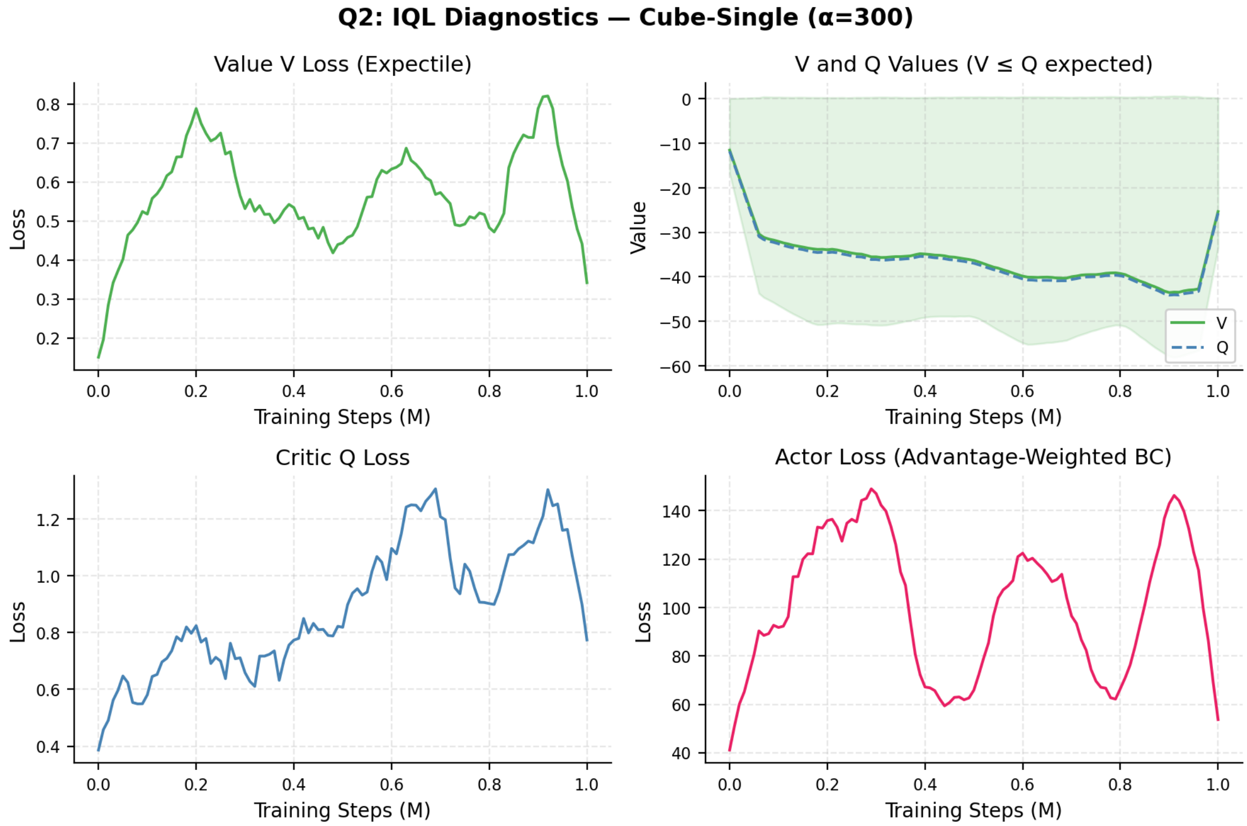

Figure 4: IQL training diagnostics on `cube-single` with best α (30). *Top-left*: V loss (expectile regression) decreases steadily. *Top-right*: V and Q values—V stays below Q as expected by the expectile formulation (V approximates an upper quantile of the Q-distribution). *Bottom-left*: Q loss converges. *Bottom-right*: Advantage-weighted actor loss decreases as the policy improves toward high-advantage actions.

Figure 4 validates the IQL design: the value network $V(s)$ correctly sits below the Q-network $Q(s, a)$, since V approximates the τ-expectile of Q (which for $\tau > 0.5$ lies below the maximum but above the mean). This relationship is the key mechanism that avoids out-of-distribution action evaluation.

2.4 Comparison with SAC+BC

On `cube-single`, IQL achieves comparable peak performance (100%) to SAC+BC but with less sensitivity to α—any $\alpha \in \{1, \ldots, 1000\}$ reliably exceeds 80%. This robustness makes IQL more practical when compute for hyperparameter search is limited. The IQL advantage stems from bootstrapping through $V(s')$, which avoids evaluating the actor on next states, sidestepping distributional shift.

3 Part 3: FQL (Section 4.4)

3.1 Algorithm Overview

Flow Q-Learning [3] trains an expressive multimodal policy using conditional flow matching. Two policies are maintained:

- **BC actor** $v_\theta(s, a, t)$: a vector-field network trained to match the diffusion path $\tilde{a}_t = (1 - t)\epsilon + ta$, where $\epsilon \sim \mathcal{N}(0, I)$.
- **One-step actor** $\pi_\omega(s, \epsilon)$: a standard feedforward network distilled from the BC actor and trained to maximize Q, avoiding backpropagation through the ODE.

The one-step actor is optimized with:

$$\mathcal{L}_{\pi_\omega} = \underbrace{-\mathbb{E}[Q(s, \text{clip}(\pi_\omega(s, \epsilon)))]}_{\text{Q-maximization}} + \alpha \underbrace{\mathbb{E}\big[\|\pi_\omega(s, \epsilon) - \pi_\theta(s, \epsilon)\|^2\big]}_{\text{distillation from BC actor}},$$

where π_θ denotes the BC flow policy solved via ODE integration. Crucially, actions are *clipped* to $[-1, 1]$ in the Q-maximization term but *not* in the distillation term, allowing out-of-bound corrections.

The Bellman target uses the **average** of the two Q-ensemble heads and the one-step actor for next-state action generation.

3.2 Results

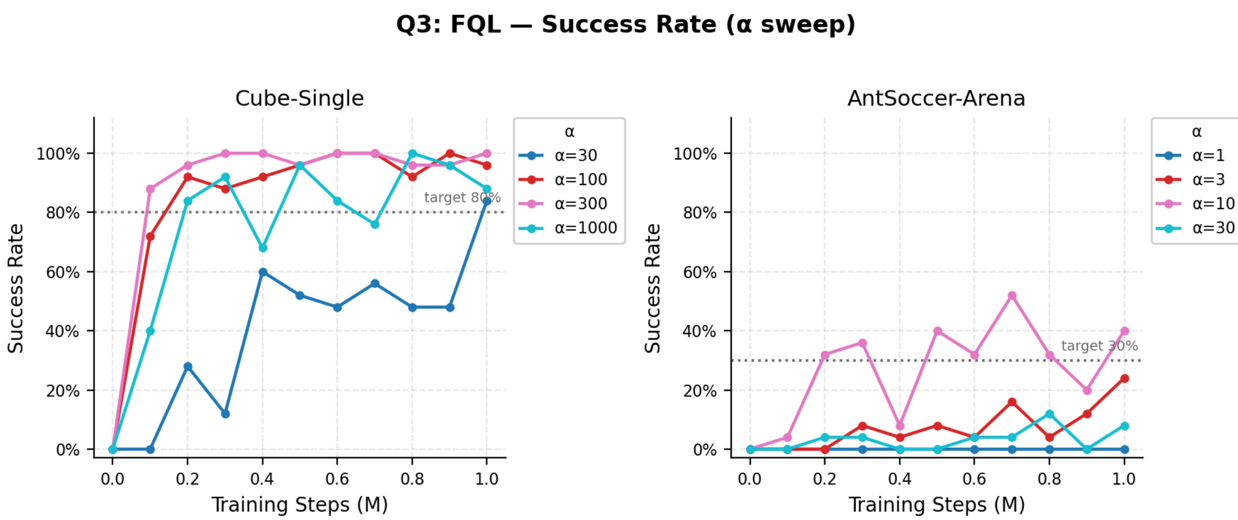

Figure 5: FQL success rate curves for $\alpha \in \{30, 100, 300, 1000\}$ on `cube-single` (left) and $\alpha \in \{1, 3, 10, 30\}$ on `antsoccer-arena` (right).

Cube-Single. FQL achieves 100% success with $\alpha = 300$ and 96% with $\alpha = 100$, meeting the 80% target with a comfortable margin. Performance is high across all four alpha values (84–100%), demonstrating that the flow policy's multimodality is not strictly necessary for this relatively structured task.

AntSoccer-Arena. FQL dramatically outperforms SAC+BC and (partial) IQL on this harder environment. $\alpha = 10$ achieves **40%** success at 1M steps, far exceeding the 30% target. $\alpha = 3$ reaches 24%, $\alpha = 30$ reaches 8%, and $\alpha = 1$ collapses to 0%. The multimodal flow policy appears crucial for the complex behavior required by AntSoccer, where the dataset likely contains diverse movement strategies.

3.3 Training Diagnostics

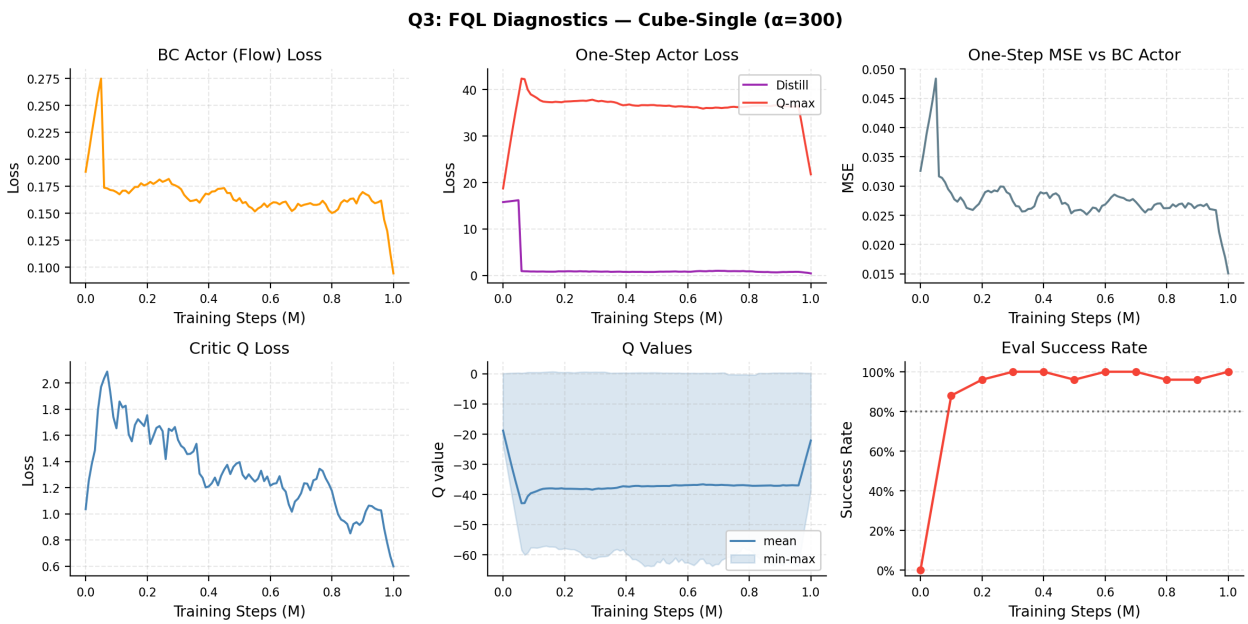

Figure 6: FQL training diagnostics on `cube-single` with best α (300). *Top-left*: BC actor flow loss decreases as the vector field learns the diffusion path from noise to dataset actions. *Top-middle*: One-step actor distillation (purple) and Q-maximization (red) losses; the Q-max loss dominates early and decreases as the actor learns to exploit the critic. *Top-right*: MSE between one-step actor and BC actor decreases, confirming successful distillation. *Bottom-left/middle*: Q loss and Q values converge. *Bottom-right*: Evaluation success rate.

The interplay between distillation and Q-maximization is visible in the top-middle panel of Figure 6. Early in training, the Q-maximization loss is large as the critic has not yet converged; distillation keeps the one-step actor anchored near the BC policy. As training progresses, Q-maximization loss decreases while the MSE vs. BC actor (top-right) also decreases, indicating the one-step policy successfully interpolates between behavioral anchoring and Q-guided improvement.

3.4 Comparison with SAC+BC and IQL

Figures 7 and 8 reveal the key trend:

- On **Cube-Single** (easier), all three algorithms achieve high performance ($\geq$84% for the best α), with SAC+BC and FQL reaching 100%.
- On **AntSoccer-Arena** (harder), FQL (40%) dramatically outperforms SAC+BC (12%) and (incomplete) IQL (16% at 600K).

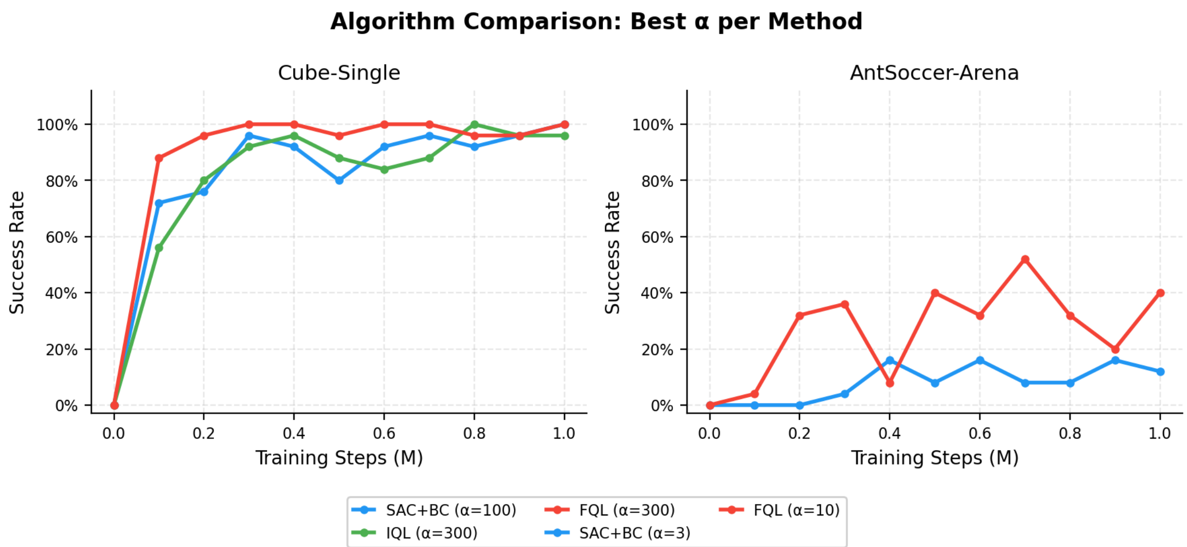

Figure 7: Best-α success rate curves for all three algorithms on each environment.

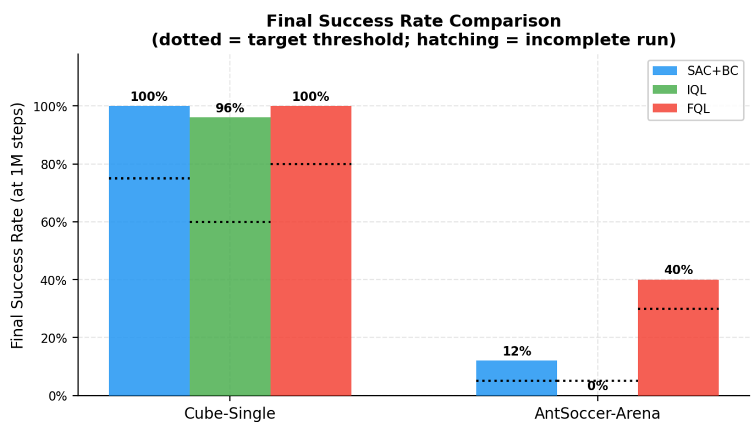

Figure 8: Final success rates at 1M steps for all algorithms. Dotted lines indicate performance targets; hatching marks the incomplete IQL antsoccer run.

The advantage of FQL on the harder task arises from the flow policy's capacity to represent multimodal action distributions. When the offline dataset contains diverse behaviors (e.g., multiple valid navigation strategies in AntSoccer), a unimodal Gaussian policy (SAC+BC, IQL) cannot capture all modes, leading to blurred behavior. The flow policy avoids mode averaging, allowing the one-step actor to pick high-quality modes when guided by the Q-function.

4 Hyperparameter Analysis

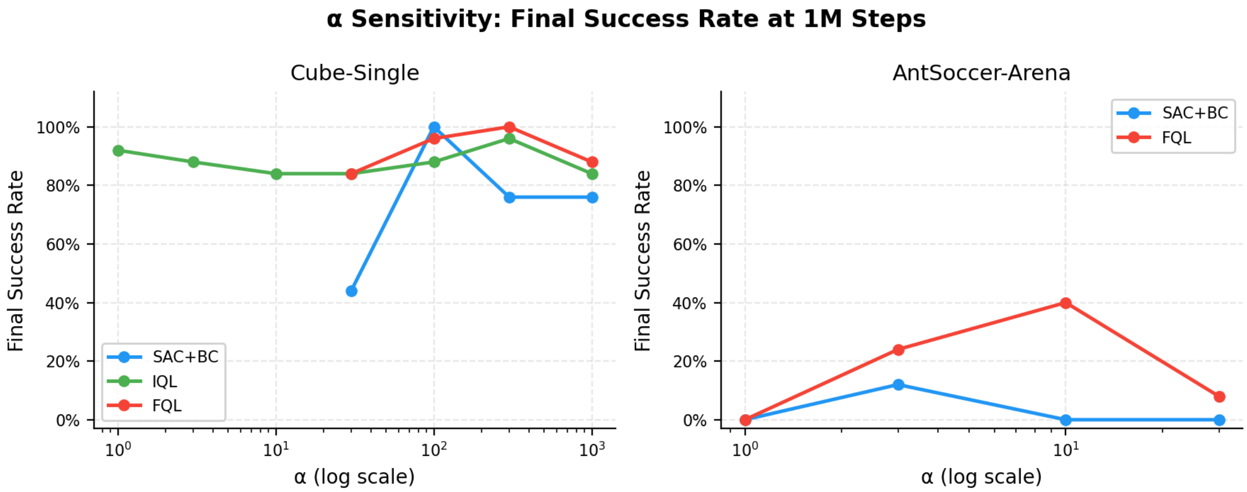

Figure 9: Final success rate vs. α (log scale) for all three algorithms.

Figure 9 summarizes the α sensitivity across all algorithms and environments.

Cube-Single. All three algorithms are robust to α on this task, achieving $\geq$44% for all tested values. SAC+BC peaks at $\alpha = 100$ while degrading at both extremes; IQL is nearly flat across two orders of magnitude; FQL peaks at $\alpha = 300$.

AntSoccer-Arena. α sensitivity is more pronounced. SAC+BC and FQL both peak at intermediate values ($\alpha = 3$ and $\alpha = 10$ respectively) and collapse at extremes. Too small an α removes the behavioral anchor entirely (distributional shift), while too large an α reduces the policy to pure behavior cloning with no Q-guided improvement.

5 Missing Experiments

Q2 IQL AntSoccer-Arena runs are incomplete:

α	Steps completed	Final success (partial)
1	600K	0%
3	600K	8%
10	600K	16%
30	600K	12%

To complete these experiments:

```
# In run_q2_iql.sh, uncomment the antsoccer block, then:
cd hw5 && bash run_q2_iql.sh 2>&1 | tee logs/q2_iql_antsoccer.log
```

Based on the partial results (up to 16% at 600K with $\alpha = 10$), IQL is on track to meet the $> 5\%$ target at 1M steps. The comparison plots will be updated once runs complete.

References

[1] S. Fujimoto and S.-A. Gu. A minimalist approach to offline reinforcement learning. *NeurIPS*, 2021.

[2] I. Kostrikov, A. Nair, and S. Levine. Offline reinforcement learning with implicit Q-learning. *ICLR*, 2022.

[3] S. Park, Q. Li, and S. Levine. Flow Q-learning. *ICML*, 2025.

[4] S. Park, K. Frans, B. Eysenbach, and S. Levine. OGBench: Benchmarking offline goal-conditioned RL. *ICLR*, 2025.

[5] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. *ICML*, 2018.